University of Centrat Punjab

Date:

Hill Cipher.

matrix 2x2 matrix 3x3.

↳ multi-cipher ↑ ↑

encrypt groups of letters digraph, trigraph, polygraph.

key: would be a matrix.

Review Algebra:

→ Matrix arithemtic modulo 26.
→ Square matrix. → Key would be a square matrix.
→ Determinant.
→ Multiplicative Inverse.

Hill Cipher → Encryption & Decryption:-

Encryption:

$$C = E(K, P) = P \times K \bmod 26.$$

Plain text x Key mod 26.
matrix matrix.

Decryption:

$$P = D(K, C) = C K^{-1} \bmod 26. = P \times K \times K^{-1} \bmod 26$$

ciphertext x Key inverse
matrix matrix.

↘ we must know about determinant.

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Hill Cipher:-

Plaint text :- HELP ME TO SOLVE

Key: $\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$

Given Text:

HEL PME TOS OLV EXX ← we are going to encrypt 3 letters at a time.

H	E	L	P	M	E	T	O	S	O	L	V	E
7	4	11	15	12	4	19	14	18	14	11	21	4

Encryption:

$C = KP \bmod 26$

Firstly we will take HEL → 7, 4, 11

$$= \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \begin{bmatrix} 7 \\ 4 \\ 11 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 6\times7 + 24\times4 + 1\times11 \\ 13\times7 + 16\times4 + 10\times11 \\ 20\times7 + 17\times4 + 15\times11 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 42 + 96 + 11 \\ 91 + 64 + 10 \\ 140 + 68 + 165 \end{bmatrix} \bmod 26$$

Since $149 \div 26 = 5.730\ldots$ we will consider integer before decimal only.

Note: $149 - (149 \div 26) \times 26$
$= 149 - 5 \times 26$
$= 149 - 130$
$= 19$

$$= \begin{bmatrix} 149 \\ 265 \\ 373 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 19 \\ 5 \\ 9 \end{bmatrix}$$

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19, 5, 9

so HEL will be encrypted as T F J

Decryption:

$P = K^{-1}C \bmod 26$.

K^{-1} is inverse of key matrix.

$$K^{-1} = \left[\frac{1}{\det K} \times \text{adj}K\right] \bmod 26.$$

$$K = \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

To solve this first we will consider row 1

$$\det K = \begin{bmatrix} +6 & -24 & +1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \bmod 26$$

(1) ignore/hide

$$= 6(16\times15 - 17\times10) - 24(13\times15 - 20\times10) + 1(13\times17 - 20\times16)$$

$$= 420 + 120 - 99$$

$$= 441 \bmod 26$$

$$\det K = 25$$

(2) ignore/hide

$$\begin{bmatrix} +6 & -24 & +1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

(3) ignore/hide

$$\begin{bmatrix} +6 & -24 & +1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

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$$\frac{1}{\det K} = \frac{1}{25}$$

$$= 25^{-1} \bmod 26$$

$$= 25$$

$\because ax \bmod m = 1$
for which value of $x = 1$
$= 25$.

Now [Method 1]

$$\text{adj } K = \begin{bmatrix} 6 & 24 & 1 & 6 & 24 \\ 13 & 16 & 10 & 13 & 16 \\ 20 & 17 & 15 & 20 & 17 \end{bmatrix}$$

here we repeat first 2 columns of key matrix.

$$\text{adj } K = \begin{vmatrix} 6 & 24 & 1 & 6 & 24 \\ 13 & 16 & 10 & 13 & 16 \\ 20 & 17 & 15 & 20 & 17 \\ 6 & 24 & 1 & 6 & 24 \\ 13 & 16 & 10 & 13 & 16 \end{vmatrix}$$

here we repeat first 2 rows.

$$= \begin{vmatrix} 16 & 10 & 13 & 16 \\ 17 & 15 & 20 & 17 \\ 24 & 1 & 6 & 24 \\ 16 & 10 & 13 & 16 \end{vmatrix}$$

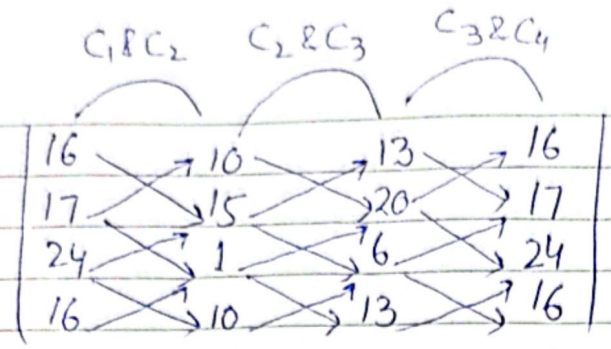
ignore first row and first column

Now

we will take first 2 columns and write result in row 1 Then
take column 2 and 3 and write result in row 2
at last
take column 3 and 4 and write result in row 3.

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$$= \begin{vmatrix} 16\times15-17\times10 & 17\times1-24\times15 & 24\times10-16\times1 \\ 10\times20-15\times13 & 15\times6-1\times20 & 1\times13-10\times6 \\ 13\times7-20\times16 & 20\times24-6\times17 & 6\times16-13\times24 \end{vmatrix}$$

$$\text{adj } K = \begin{vmatrix} 18 & 21 & 16 \\ 5 & 18 & 5 \\ 5 & 14 & 18 \end{vmatrix}$$

Now

$$K^{-1} = \left[\frac{1}{\det K} \times \text{adj } K\right] \bmod 26$$

$$= 25 \times \begin{bmatrix} 18 & 21 & 16 \\ 5 & 18 & 5 \\ 5 & 14 & 18 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 450 & 525 & 400 \\ 125 & 450 & 125 \\ 125 & 350 & 450 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{bmatrix}$$

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Method [2]

$$\text{adj } K = \begin{bmatrix} 16\times15 - 17\times10 & 13\times15 - 20\times10 & 13\times17 - 20\times16 \\ 24\times15 - 17\times1 & 6\times15 - 20\times1 & 6\times17 - 20\times24 \\ 24\times10 - 16\times1 & 6\times10 - 13\times1 & 6\times16 - 13\times24 \end{bmatrix}$$

$$= \begin{bmatrix} 70 & -5 & -99 \\ 343 & 70 & -378 \\ 224 & 47 & 96-312 = -216 \end{bmatrix}$$

now applying $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$ on above :-

$$= \begin{bmatrix} 70 & 5 & -99 \\ -343 & 70 & 378 \\ 224 & -47 & -216 \end{bmatrix}$$

now for converting negative terms into positive ones. (we) have :-

$-99 + 26n = 5$ where $n = 4$
$-343 + 26n = 21$ where $n = 14$
$-47 + 26n = 5$ where $n = 2$
$-216 + 26n = 18$ where $n = 9$.

$$\text{adj}K = \begin{bmatrix} 70 & 5 & 5 \\ 21 & 70 & 378 \\ 224 & 5 & 18 \end{bmatrix}$$

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Now taking transpose :- [In case of 2x2 we will not take transpose.

$$= \begin{bmatrix} 70 & 21 & 224 \\ 5 & 70 & 5 \\ 5 & 378 & 18 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 18 & 21 & 16 \\ 5 & 18 & 5 \\ 5 & 14 & 18 \end{bmatrix}$$

same adj with method 1

Now

$$K^{-1} = \left[\frac{1}{\det K} \times \text{adj } K \right] \text{ mod } 26$$

$$= 25 \begin{bmatrix} 18 & 21 & 16 \\ 5 & 18 & 5 \\ 5 & 14 & 18 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 450 & 525 & 400 \\ 125 & 450 & 125 \\ 125 & 350 & 450 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{bmatrix}$$

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Now for decryption we have:-

$$P = K^{-1} C \bmod 26.$$

$$= \begin{bmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{bmatrix} \begin{bmatrix} 19 \\ 5 \\ 9 \end{bmatrix} \bmod 26.$$

$$= \begin{bmatrix} 8\times19 + 5\times5 + 10\times9 \\ 21\times19 + 8\times5 + 21\times9 \\ 21\times19 + 12\times5 + 8\times9 \end{bmatrix} \bmod 26.$$

$$= \begin{bmatrix} 152 + 25 + 90 \\ 399 + 40 + 189 \\ 399 + 60 + 72 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 267 \\ 628 \\ 531 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 7 \\ 4 \\ 11 \end{bmatrix}$$

So T F J is decrypted as 7 4 11 which is H E L ← same as the given text.

Note: Same method will be applied for remaining letters of given text.

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